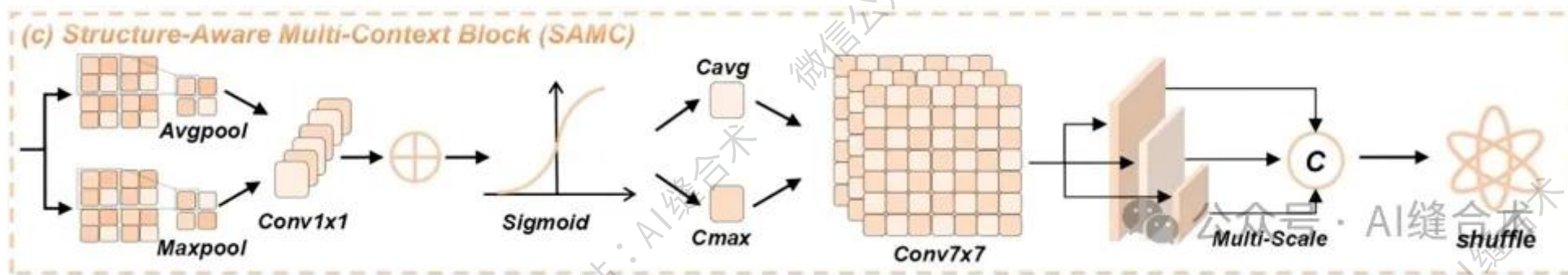


核心速览



论文题目: SEMC: Structure-Enhanced Mixture-of-Experts Contrastive Learning for Ultrasound Standard Plane Recognition

中文题目: SEMC : 结构增强的专家混合对比学习用于超声标准平面识别

论文链接: <https://arxiv.org/pdf/2511.12559>

所属单位: 中国海洋大学信息科学与工程学院, 香港理工大学健康科技与信息学系, 山东大学信息科学与工程学院

核心速览: 提出 SEMC 框架, 通过语义-结构融合模块和混合专家对比识别模块, 结合多尺度结构信息与专家引导对比学习, 提升超声标准平面识别性能, 并构建 LP2025 肝脏超声数据集。

论文内容详细解读:

<https://mp.weixin.qq.com/s/O1UQSkOjQZaZRwJY8zd43A>

```

SAMC(
  (cab): CAB(
    (avg_pool): AdaptiveAvgPool2d(output_size=1)
    (max_pool): AdaptiveMaxPool2d(output_size=1)
    (activation): ReLU(inplace=True)
    (fc1): Conv2d(3, 1, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (fc2): Conv2d(1, 3, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (sigmoid): Sigmoid()
  )
  (sab): SAB(
    (conv): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
    (sigmoid): Sigmoid()
  )
  (msconv): MSConv(
    (branches): ModuleList(
      (0): Sequential(
        (0): Conv2d(3, 3, kernel_size=(1, 1), stride=(1, 1))
        (1): BatchNorm2d(3, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (2): ReLU(inplace=True)
      )
      (1): Sequential(
        (0): Conv2d(3, 3, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
        (1): BatchNorm2d(3, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (2): ReLU(inplace=True)
      )
      (2): Sequential(
        (0): Conv2d(3, 3, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2))
        (1): BatchNorm2d(3, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (2): ReLU(inplace=True)
      )
    )
    (fusion): Conv2d(9, 3, kernel_size=(1, 1), stride=(1, 1))
  )
)
input_tensor_shape : torch.Size([1, 3, 256, 256])
output_tensor_shape : torch.Size([1, 3, 256, 256])

```

哔哩哔哩/微信公众号：AI缝合术，独家整理！